

## 6.3 Permutations and Sign

An alternating form changes sign when two of its arguments are swapped (Theorem 6.2.1). To compare  $f(\mathbf{v}_3, \mathbf{v}_1, \mathbf{v}_2)$  with  $f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ , we reach one list from the other by swaps and collect a factor  $-1$  per swap. But the same rearrangement can be reached by different sequences of swaps, and Chapter 0 showed that even their number is not unique. If one route used an even number of swaps and another an odd number, we would get  $f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = -f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$  for every alternating form, which is absurd for signed volume. This section rules that out. It attaches to each permutation a sign defined without any choices, proves that the sign is multiplicative, and deduces that the parity of the number of transpositions is well defined, as Chapter 0 promised.

### 6.3.1 Permutations, recalled

Recall from Chapter 0 that  $S_n$  is the group of bijections  $\{1, \dots, n\} \rightarrow \{1, \dots, n\}$ , with  $n!$  elements. A permutation can be written in two-line notation, with  $\sigma(i)$  below  $i$ , or as a product of cycles. The product  $\sigma\tau$  is the composition  $\sigma \circ \tau$ : **first**  $\tau$ , **then**  $\sigma$ , so products of cycles are read from right to left (Example (Computing with cycles)). A transposition  $(i\ j)$  swaps  $i$  and  $j$  and fixes everything else, and for  $n \geq 2$  every permutation is a product of transpositions (Theorem 0.10.4). The product is far from unique, since

$$(1\ 2\ 3) = (1\ 3)(1\ 2) = (1\ 2)(2\ 3) = (1\ 3)(1\ 2)(1\ 2)(1\ 2).$$

Here the numbers of transpositions are 2, 2, 4, all even. The goal is to show this is no accident.

Why can we not simply **define** the sign of  $\sigma$  as  $(-1)^r$ , where  $\sigma$  is a product of  $r$  transpositions? Because that definition depends on a choice of product, and we would have to prove that every choice gives the same answer. That is the statement we are trying to prove. We need a definition that involves no choice at all.

### 6.3.2 Inversions and the sign

To measure how much a permutation scrambles  $1, 2, \dots, n$ , look at the bottom row  $\sigma(1), \sigma(2), \dots, \sigma(n)$  of its two-line notation and count the pairs of entries that are in the wrong order. A sorted row has none, and each swap of two neighboring entries changes the count by exactly one. So the count is a natural candidate for tracking the parity of swaps.

*The sign of a permutation records whether it puts an even or an odd number of pairs out of order.*

#### Definition 6.3.1 — Inversion

Let  $\sigma \in S_n$ . An **inversion** of  $\sigma$  is a pair  $(i, j)$  of positions with  $1 \leq i < j \leq n$  and  $\sigma(i) > \sigma(j)$ . We write  $\text{inv}(\sigma)$  for the number of inversions of  $\sigma$ .

#### Definition 6.3.2 — Sign of a Permutation

The **sign** of  $\sigma \in S_n$  is

$$\operatorname{sgn} \sigma = (-1)^{\operatorname{inv}(\sigma)} \in \{1, -1\}.$$

The permutation  $\sigma$  is **even** if  $\operatorname{sgn} \sigma = 1$  and **odd** if  $\operatorname{sgn} \sigma = -1$ .

In words: an inversion is a pair of positions whose order  $\sigma$  **reverses**. The condition  $i < j$  makes each pair of positions count once, and  $\sigma(i) > \sigma(j)$  says the values come out in the opposite order. Since  $\operatorname{inv}(\sigma)$  is the size of a subset of the finite set of pairs  $i < j$ , it is a well-defined integer between 0 and  $n(n-1)/2$ , and there is nothing to choose. That is the whole reason for this definition.

**The identity.** Its bottom row is sorted, so  $\operatorname{inv}(\operatorname{id}) = 0$  and  $\operatorname{id}$  is even. In  $S_1 = \{\operatorname{id}\}$  this is the only permutation, with sign 1. The degenerate case matters: the identity is the empty product of transpositions, and its sign  $1 = (-1)^0$  is what a product of 0 swaps should give.

**All of  $S_3$ .** Here is each permutation with its bottom row and inversions.

| $\sigma$   | id    | (1 2)  | (2 3)  | (1 3)                  | (1 2 3)        | (1 3 2)        |
|------------|-------|--------|--------|------------------------|----------------|----------------|
| bottom row | 1 2 3 | 2 1 3  | 1 3 2  | 3 2 1                  | 2 3 1          | 3 1 2          |
| inversions | none  | (1, 2) | (2, 3) | (1, 2), (1, 3), (2, 3) | (1, 2), (2, 3) | (1, 2), (1, 3) |
| sgn        | 1     | -1     | -1     | -1                     | 1              | 1              |

For (1 2 3), the bottom row is  $\sigma(1), \sigma(2), \sigma(3) = 2, 3, 1$ , and the out-of-order pairs of positions are (1, 3), since  $2 > 1$ , and (2, 3), since  $3 > 1$ .

Two things stand out. The three transpositions are odd, although (1 3) has three inversions, not one. And the two 3-cycles are even, consistent with  $(1 2 3) = (1 3)(1 2)$  being a product of two transpositions.

**Why all pairs, not just neighbors.** A tempting shortcut counts only **adjacent** positions  $(i, i+1)$  with  $\sigma(i) > \sigma(i+1)$ . For (1 3), with bottom row 3, 2, 1, this count is 2, which would make (1 3) “even”. But  $(1 3) = (1 2)(2 3)(1 2)$ , as one checks by following  $1 \mapsto 2 \mapsto 3 \mapsto 3$ ,  $2 \mapsto 1 \mapsto 1 \mapsto 2$ ,  $3 \mapsto 3 \mapsto 2 \mapsto 1$ , and each factor has adjacent count 1. A count that is to be multiplicative must give  $(-1)^3 = -1$  here, so the adjacent count is the wrong one.

*Quick check.* Find  $\operatorname{sgn}(1\ 4)$  in  $S_4$  by counting inversions.

*Solution.* The bottom row is 4, 2, 3, 1. The pairs of positions in the wrong order are (1, 2), (1, 3), (1, 4), since 4 is larger than everything after it, and (2, 4), (3, 4), since  $2 > 1$  and  $3 > 1$ . The pair (2, 3) is in order. So  $\operatorname{inv}(1\ 4) = 5$  and  $\operatorname{sgn}(1\ 4) = -1$ .

### 6.3.3 The sign is multiplicative

The key property is that signs multiply when permutations are composed.

#### Theorem 6.3.1 — The Sign Is Multiplicative

For all  $\sigma, \tau \in S_n$ ,

$$\operatorname{sgn}(\sigma\tau) = \operatorname{sgn} \sigma \cdot \operatorname{sgn} \tau.$$

*Idea.* Count inversions one pair at a time, and record each pair as  $+1$  (order kept) or  $-1$  (order reversed). The sign is the product of these  $\pm 1$ 's. Applying  $\tau$  and then  $\sigma$  moves a pair of positions  $\{i, j\}$  to  $\{\tau(i), \tau(j)\}$  and then to  $\{\sigma\tau(i), \sigma\tau(j)\}$ . The order is reversed overall exactly when it is reversed at one of the two stages but not both, which is multiplication of  $\pm 1$ 's. Finally, as  $\{i, j\}$  runs over all pairs, so does  $\{\tau(i), \tau(j)\}$ , so the contributions of  $\sigma$  multiply to  $\operatorname{sgn} \sigma$ , just in a different order.

*Proof.* For a non-zero integer  $m$  let  $s(m) = 1$  if  $m > 0$  and  $s(m) = -1$  if  $m < 0$ . Then  $s(mm') = s(m)s(m')$  for non-zero integers  $m, m'$ , by the sign rules for products of integers.

Let  $\mathcal{P}$  be the set of 2-element subsets  $\{i, j\}$  of  $\{1, \dots, n\}$ . For  $\rho \in S_n$  and  $\{i, j\} \in \mathcal{P}$ , define

$$\varepsilon_\rho(\{i, j\}) = s((\rho(i) - \rho(j))(i - j)).$$

This is defined:  $i \neq j$  and  $\rho(i) \neq \rho(j)$  since  $\rho$  is injective, so the product is non-zero; and exchanging the names  $i$  and  $j$  negates both factors, so the product does not depend on the order in which the subset is written. If  $i < j$ , then  $i - j < 0$ , so  $\varepsilon_\rho(\{i, j\}) = -1$  exactly when  $\rho(i) > \rho(j)$ , that is, when  $(i, j)$  is an inversion of  $\rho$ . Hence, by [Definition 6.3.2](#),

$$\operatorname{sgn} \rho = \prod_{P \in \mathcal{P}} \varepsilon_\rho(P). \quad (1)$$

Now let  $\{i, j\} \in \mathcal{P}$ . Since  $\tau(i) \neq \tau(j)$ , we have  $(\tau(i) - \tau(j))^2 > 0$ . Multiplying inside  $s$  by this positive integer does not change the value, so

$$\varepsilon_{\sigma\tau}(\{i, j\}) = s\left((\sigma(\tau(i)) - \sigma(\tau(j))) (\tau(i) - \tau(j)) \cdot (\tau(i) - \tau(j)) (i - j)\right) = \varepsilon_\sigma(\{\tau(i), \tau(j)\}) \varepsilon_\tau(\{i, j\}).$$

Let  $\hat{\tau}: \mathcal{P} \rightarrow \mathcal{P}$ ,  $\{i, j\} \mapsto \{\tau(i), \tau(j)\}$ ; the image has two elements because  $\tau$  is injective. The same construction with  $\tau^{-1}$  gives an inverse function, so  $\hat{\tau}$  is a bijection ([Theorem 0.4.7](#)). Taking the product over all  $P \in \mathcal{P}$  and using (1) three times,

$$\operatorname{sgn}(\sigma\tau) = \prod_{P \in \mathcal{P}} \varepsilon_\sigma(\hat{\tau}(P)) \cdot \prod_{P \in \mathcal{P}} \varepsilon_\tau(P) = \prod_{Q \in \mathcal{P}} \varepsilon_\sigma(Q) \cdot \operatorname{sgn} \tau = \operatorname{sgn} \sigma \cdot \operatorname{sgn} \tau,$$

where the second equality substitutes  $Q = \hat{\tau}(P)$ , which runs over  $\mathcal{P}$  exactly once because  $\hat{\tau}$  is a bijection, and reorders the finite product of integers. This proves the theorem. ■

By induction on the number of factors,  $\operatorname{sgn}(\sigma_1\sigma_2 \cdots \sigma_r) = \operatorname{sgn} \sigma_1 \cdots \operatorname{sgn} \sigma_r$ . In the language of Chapter 0, the set  $\{1, -1\}$  is a group under multiplication (it contains 1, and it is closed under products and inverses, since  $(-1)^{-1} = -1$ ), and the theorem says exactly that

$$\operatorname{sgn}: S_n \rightarrow \{1, -1\}$$

is a group homomorphism ([Definition 0.10.3](#)).

## 6.3.4 Transpositions and parity

Now we can compute the sign of the building blocks.

### Corollary 6.3.2 — Sign of a Transposition

Every transposition is odd:  $\text{sgn}(i\ j) = -1$  for all  $i \neq j$  in  $\{1, \dots, n\}$ .

*Proof.* Since  $(i\ j) = (j\ i)$ , we may assume  $i < j$ . Let  $\tau = (i\ j)$  and let  $p < q$  be positions. We decide when  $\tau(p) > \tau(q)$ .

- If neither  $p$  nor  $q$  lies in  $\{i, j\}$ , then  $\tau(p) = p < q = \tau(q)$ : no inversion.
- If  $(p, q) = (i, j)$ , then  $\tau(i) = j > i = \tau(j)$ : one inversion.
- If  $p = i$  and  $q \neq j$ , then  $\tau(p) = j$  and  $\tau(q) = q$ , an inversion exactly when  $i < q < j$ . This gives  $j - i - 1$  inversions.
- If  $q = j$  and  $p \neq i$ , then  $\tau(p) = p$  and  $\tau(q) = i$ , an inversion exactly when  $i < p < j$ . This gives  $j - i - 1$  inversions.
- If  $p = j$  (so  $q > j$ ), then  $\tau(p) = i < q = \tau(q)$ ; if  $q = i$  (so  $p < i$ ), then  $\tau(p) = p < j = \tau(q)$ : no inversion.

These cases cover every pair, so  $\text{inv}(\tau) = 1 + 2(j - i - 1)$ , an odd number. Hence  $\text{sgn } \tau = -1$ . ■

The Quick check above is the case  $(i, j) = (1, 4)$ :  $1 + 2 \cdot 2 = 5$  inversions. Combining the corollary with multiplicativity gives the fact that Chapter 0 left open.

### Corollary 6.3.3 — Parity Is Well Defined

Let  $n \geq 2$  and  $\sigma \in S_n$ . If  $\sigma = \tau_1 \cdots \tau_r$  is a product of  $r$  transpositions, then  $\text{sgn } \sigma = (-1)^r$ . Consequently, if also  $\sigma = \tau'_1 \cdots \tau'_s$  with transpositions  $\tau'_m$ , then  $r$  and  $s$  are both even or both odd.

*Proof.* By [Theorem 6.3.1](#), extended to  $r$  factors, and [Corollary 6.3.2](#),  $\text{sgn } \sigma = \text{sgn } \tau_1 \cdots \text{sgn } \tau_r = (-1)^r$ . In the same way  $\text{sgn } \sigma = (-1)^s$ . Hence  $(-1)^r = (-1)^s$ , so  $r - s$  is even. ■

So a permutation is even exactly when it is a product of an even number of transpositions, and the “number of swaps” has a well-defined parity even though the number itself is not defined.

### Proposition 6.3.4 — Sign of a Cycle

A  $k$ -cycle has sign  $(-1)^{k-1}$ . In particular, cycles of even length are odd and cycles of odd length are even.

*Proof.* For  $k = 1$  the cycle is id, with sign  $1 = (-1)^0$ . Let  $k \geq 2$  and let  $\gamma = (i_1 i_2 \cdots i_k)$ . We claim

$$\gamma = (i_1 i_k)(i_1 i_{k-1}) \cdots (i_1 i_3)(i_1 i_2).$$

Apply the factors from right to left. The element  $i_1$  is sent to  $i_2$  by  $(i_1 i_2)$ , and no later factor involves  $i_2$ . For  $2 \leq m \leq k-1$ , the element  $i_m$  is fixed by the factors before  $(i_1 i_m)$ , sent to  $i_1$  by it, then to  $i_{m+1}$  by  $(i_1 i_{m+1})$ , and no later factor involves  $i_{m+1}$ . The element  $i_k$  is fixed until the last factor  $(i_1 i_k)$  sends it to  $i_1$ . Elements outside the cycle are fixed by every factor. This agrees with  $\gamma$  everywhere, proving the claim. The product has  $k-1$  transpositions, so  $\text{sgn } \gamma = (-1)^{k-1}$  by [Corollary 6.3.3](#). ■

**Warning.** Only the parity of the number of transpositions is determined, and it is **not** the parity of the cycle length. The identity in  $S_3$  equals  $(1\ 2)(1\ 2)$  and also  $(1\ 2)(1\ 3)(1\ 2)(1\ 3)(1\ 2)(1\ 3)$ , since  $((1\ 2)(1\ 3))^3 = (1\ 3\ 2)^3 = \text{id}$ ; the numbers 2 and 6 differ, only their parity agrees. And a 3-cycle, “of length 3”, is **even**:  $(1\ 2\ 3) = (1\ 3)(1\ 2)$ . One more trap concerns order:  $(1\ 3)(1\ 2) = (1\ 2\ 3)$  but  $(1\ 2)(1\ 3) = (1\ 3\ 2)$ . Multiplying in the wrong order gives the wrong permutation and the wrong bottom row, even though, by [Theorem 6.3.1](#), it happens to give the right sign.

## 6.3.5 The alternating group

Since  $\text{sgn}$  is a homomorphism, its kernel is a subgroup of  $S_n$  (Theorem 0.10.3).

### Definition 6.3.3 — Alternating Group

The **alternating group**  $A_n$  is the kernel of  $\text{sgn}: S_n \rightarrow \{1, -1\}$ , that is, the subgroup of **even** permutations in  $S_n$ .

For example,  $A_3 = \{\text{id}, (1\ 2\ 3), (1\ 3\ 2)\}$  by the table above, and  $A_2 = \{\text{id}\}$ . For  $n \geq 2$ , exactly half of the permutations are even (Exercise C1). The odd permutations do **not** form a subgroup: they do not contain id. The name is no coincidence: as we see below and in the next section, rearranging the arguments of an alternating form by an even permutation leaves its value unchanged.

## 6.3.6 Back to alternating forms

Here is why this section exists. Let  $f$  be an alternating 3-linear form and compare  $f(\mathbf{v}_3, \mathbf{v}_1, \mathbf{v}_2)$  with  $f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ . Swapping the first two arguments of  $(\mathbf{v}_3, \mathbf{v}_1, \mathbf{v}_2)$  gives  $(\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_2)$ , and swapping the last two then gives  $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ . By Theorem 6.2.1 (a) each swap costs a factor  $-1$ , so  $f(\mathbf{v}_3, \mathbf{v}_1, \mathbf{v}_2) = (-1)^2 f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ . The list  $(\mathbf{v}_3, \mathbf{v}_1, \mathbf{v}_2)$  is  $(\mathbf{v}_{\sigma(1)}, \mathbf{v}_{\sigma(2)}, \mathbf{v}_{\sigma(3)})$  for  $\sigma = (1\ 3\ 2)$ , with bottom row 3, 1, 2, and the factor 1 we found is  $\text{sgn } \sigma$ .

A different sequence of swaps could have been used, possibly a longer one. By [Corollary 6.3.3](#), every such sequence has an even number of swaps, so every route gives the same factor. In general, the next section proves that  $f(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(k)}) = \text{sgn } \sigma \cdot f(\mathbf{v}_1, \dots, \mathbf{v}_k)$  for every alternating  $k$ -linear form, and uses it to turn each of the  $n!$  surviving terms of the expansion sketched in

the first section of this chapter into  $\pm D(\mathbf{e}_1, \dots, \mathbf{e}_n)$ . Without the corollary, the factor computed swap by swap could depend on the route, and could not be identified with  $\text{sgn } \sigma$ .

### 6.3.7 Computing signs in practice

There are now two independent ways to find a sign: count inversions, or factor into cycles and use [Proposition 6.3.4](#) with multiplicativity. Doing both is a good check.

#### Example. (A Sign Computed Two Ways)

Find the sign of the permutation  $\sigma \in S_6$  with two-line notation  $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 5 & 3 & 6 & 2 & 1 \end{pmatrix}$ , first by counting inversions and then from its cycles.

*Solution. Inversions.* Go through the bottom row 4, 5, 3, 6, 2, 1 from left to right, and for each entry count the **later** entries that are smaller:

| position $i$      | 1       | 2       | 3    | 4    | 5 | 6    |
|-------------------|---------|---------|------|------|---|------|
| $\sigma(i)$       | 4       | 5       | 3    | 6    | 2 | 1    |
| later and smaller | 3, 2, 1 | 3, 2, 1 | 2, 1 | 2, 1 | 1 | none |
| count             | 3       | 3       | 2    | 2    | 1 | 0    |

So  $\text{inv}(\sigma) = 3 + 3 + 2 + 2 + 1 + 0 = 11$ , which is odd, and  $\text{sgn } \sigma = -1$ .

*Cycles.* Following elements as in Example (Computing with cycles):  $1 \mapsto 4 \mapsto 6 \mapsto 1$ , then  $2 \mapsto 5 \mapsto 2$ , and  $3 \mapsto 3$ . So  $\sigma = (1\ 4\ 6)(2\ 5)$ . By [Proposition 6.3.4](#), the 3-cycle has sign 1 and the transposition has sign  $-1$ , so by [Theorem 6.3.1](#),  $\text{sgn } \sigma = 1 \cdot (-1) = -1$ . The two methods agree.

The following cell checks the example, and also tests [Theorem 6.3.1](#) and [Proposition 6.3.4](#) on all of  $S_5$  by brute force. Permutations are stored as tuples of values  $(\sigma(1), \dots, \sigma(n))$ , shifted to start at 0.

```
from itertools import permutations, combinations

def inv(s):
    return sum(1 for i, j in combinations(range(len(s)), 2) if s[i] > s[j])

def sgn(s):
    return (-1) ** inv(s)

def compose(s, t):
    # (s t)(i) = s(t(i)): first t, then s
    return tuple(s[t[i]] for i in range(len(t)))

def cycle(entries, n):
    # the cycle (entries[0] entries[1] ...), 0-based
    s = list(range(n))
    for a, b in zip(entries, entries[1:] + entries[:1]):
        s[a] = b
    return tuple(s)

sigma = (4, 5, 3, 6, 2, 1)
```

```

sigma0 = tuple(x - 1 for x in sigma)
print("inversions:", inv(sigma0), " sign:", sgn(sigma0))
print("(1 4 6)(2 5) equals sigma:", compose(cycle([0, 3, 5], 6), cycle([1, 4], 6))
      == sigma0)

S5 = list(permutations(range(5)))
print("multiplicative on S_5:", all(sgn(compose(s, t)) == sgn(s) * sgn(t) for s in
      S5 for t in S5))
print("k-cycles have sign (-1)^(k-1):", all(sgn(cycle(list(range(k))), 5)) == (-1) **
      (k - 1) for k in range(1, 6)))

```

[Run this code online](#)

## 6.3.8 Exercises

### 6.3.8.1 A. Check your understanding

#### Exercise A1.

- Define an inversion of  $\sigma \in S_n$  and the sign  $\text{sgn } \sigma$ .
- True or false: every 3-cycle is odd. Justify your answer.
- True or false: some permutation is a product of 5 transpositions and also a product of 8 transpositions. Justify your answer.
- What is the kernel of  $\text{sgn}: S_n \rightarrow \{1, -1\}$ ? Is the set of odd permutations a subgroup?
- True or false:  $\text{sgn}(\sigma\tau) = \text{sgn}(\tau\sigma)$  for all  $\sigma, \tau \in S_n$ , even when  $\sigma\tau \neq \tau\sigma$ . Justify your answer.
- Let  $f$  be an alternating 3-linear form. Express  $f(\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_1)$  in terms of  $f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ .

*Solution.*

- An inversion of  $\sigma$  is a pair  $(i, j)$  with  $1 \leq i < j \leq n$  and  $\sigma(i) > \sigma(j)$  ([Definition 6.3.1](#));  $\text{sgn } \sigma = (-1)^{\text{inv}(\sigma)}$  ([Definition 6.3.2](#)).
- False. By [Proposition 6.3.4](#) a 3-cycle has sign  $(-1)^2 = 1$ , so it is even; for instance  $(1\ 2\ 3) = (1\ 3)(1\ 2)$ .
- False. By [Corollary 6.3.3](#) the numbers of transpositions in two such products have the same parity, and 5 and 8 do not.
- The kernel is the alternating group  $A_n$  of even permutations ([Definition 6.3.3](#)). The odd permutations do not form a subgroup:  $\text{id}$  is even, so it is not among them.
- True. By [Theorem 6.3.1](#) both sides equal  $\text{sgn } \sigma \cdot \text{sgn } \tau$ , and multiplication of integers is commutative. For example  $(1\ 3)(1\ 2) \neq (1\ 2)(1\ 3)$ , but both are 3-cycles, with sign 1.
- Swapping the first two arguments of  $(\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_1)$  gives  $(\mathbf{v}_3, \mathbf{v}_2, \mathbf{v}_1)$ , and swapping the first and third then gives  $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ . Each swap costs a factor  $-1$  by [Theorem 6.2.1 \(a\)](#), so  $f(\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_1) = f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ . This matches the sign of the rearrangement  $\sigma$  with bottom row 2, 3, 1, namely  $\sigma = (1\ 2\ 3)$ , which is even.

### 6.3.8.2 B. Practice

#### Exercise B1: Computing Signs.

Find the sign of each permutation, both by counting inversions and by using cycles.

- (a)  $\sigma \in S_4$  with two-line notation  $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix}$ .
- (b)  $(1\ 2)(3\ 4\ 5) \in S_5$ .
- (c)  $(1\ 2\ 3)(2\ 3\ 4) \in S_4$ .

*Solution.*

- (a) *Inversions.* In the bottom row 2, 4, 1, 3, the entry 2 is followed by the smaller entry 1, the entry 4 by 1 and 3, and the entry 1 by nothing smaller. So  $\text{inv}(\sigma) = 1 + 2 + 0 = 3$  and  $\text{sgn } \sigma = -1$ . *Cycles.*  $1 \mapsto 2 \mapsto 4 \mapsto 3 \mapsto 1$ , so  $\sigma = (1\ 2\ 4\ 3)$ , a 4-cycle, with sign  $(-1)^3 = -1$  by [Proposition 6.3.4](#).
- (b) *Cycles.* By [Proposition 6.3.4](#) and [Theorem 6.3.1](#), the sign is  $(-1) \cdot (-1)^2 = -1$ . *Inversions.* The permutation sends  $1 \mapsto 2$ ,  $2 \mapsto 1$ ,  $3 \mapsto 4$ ,  $4 \mapsto 5$ ,  $5 \mapsto 3$ , so the bottom row is 2, 1, 4, 5, 3. The inversions are (1, 2), (3, 5) and (4, 5), so there are 3, and the sign is  $-1$ .
- (c) *Cycles.* Both factors are 3-cycles, so the sign is  $1 \cdot 1 = 1$ . *Inversions.* Apply (2 3 4) first:  $1 \mapsto 1 \mapsto 2$ ,  $2 \mapsto 3 \mapsto 1$ ,  $3 \mapsto 4 \mapsto 4$ ,  $4 \mapsto 2 \mapsto 3$ . The bottom row is 2, 1, 4, 3, with inversions (1, 2) and (3, 4). So the sign is  $(-1)^2 = 1$ , and in fact the product is  $(1\ 2)(3\ 4)$ .

#### Exercise B2: Sign of the Inverse.

Prove that  $\text{sgn}(\sigma^{-1}) = \text{sgn } \sigma$  for every  $\sigma \in S_n$ .

*Solution.* Let  $\sigma \in S_n$ . By [Theorem 6.3.1](#),  $\text{sgn } \sigma \cdot \text{sgn}(\sigma^{-1}) = \text{sgn}(\sigma\sigma^{-1}) = \text{sgn}(\text{id}) = 1$ , since  $\text{id}$  has no inversions. Since  $\text{sgn } \sigma \in \{1, -1\}$  and  $(\pm 1)^{-1} = \pm 1$ , multiplying both sides by  $\text{sgn } \sigma$  gives  $\text{sgn}(\sigma^{-1}) = \text{sgn } \sigma$ .

#### Exercise B3: The Reversal Permutation.

Let  $\rho \in S_n$  be the reversal,  $\rho(i) = n + 1 - i$ . Show that  $\text{inv}(\rho) = n(n-1)/2$ . Hence find the  $n$  for which  $\rho$  is even, and decide the sign of  $\rho$  for  $n = 6$  and  $n = 9$ .

*Solution.* For  $1 \leq i < j \leq n$ ,  $\rho(i) = n + 1 - i > n + 1 - j = \rho(j)$ . So **every** pair  $i < j$  is an inversion, and  $\text{inv}(\rho)$  is the number of 2-element subsets of  $\{1, \dots, n\}$ , which is  $n(n-1)/2$ . Hence  $\text{sgn } \rho = (-1)^{n(n-1)/2}$ .

Write  $n = 4q + t$  with  $t \in \{0, 1, 2, 3\}$ . Then  $n(n-1)/2$  is  $2q(4q-1)$  for  $t = 0$ ,  $2q(4q+1)$  for  $t = 1$ ,  $(4q+1)(2q+1)$  for  $t = 2$ , and  $(2q+1)(4q+3)$  for  $t = 3$ . The first two are even and the last two are odd. So  $\rho$  is even exactly when  $n$  leaves remainder 0 or 1 on division by 4. For  $n = 6$ ,  $\text{inv}(\rho) = 15$  and  $\rho$  is odd; for  $n = 9$ ,  $\text{inv}(\rho) = 36$  and  $\rho$  is even.



## 6.3.8.3 C. Going deeper

**Exercise C1: Half the Permutations Are Even.**

Let  $n \geq 2$ . Prove that exactly  $n!/2$  permutations in  $S_n$  are even.

*Hint: consider  $\sigma \mapsto (1\ 2)\sigma$ .*

*Solution.* Let  $E$  and  $O$  be the sets of even and odd permutations; they are disjoint and  $E \cup O = S_n$ , so  $|E| + |O| = n!$ . Let  $\tau = (1\ 2)$ , which exists since  $n \geq 2$ , and define  $L: S_n \rightarrow S_n$ ,  $L(\sigma) = \tau\sigma$ . By [Theorem 6.3.1](#) and [Corollary 6.3.2](#),  $\text{sgn } L(\sigma) = -\text{sgn } \sigma$ , so  $L$  maps  $E$  into  $O$  and  $O$  into  $E$ . Since  $\tau\tau = \text{id}$ ,  $L(L(\sigma)) = \sigma$ , so  $L$  restricted to  $E$  is a bijection  $E \rightarrow O$  with inverse  $L$  restricted to  $O$  ([Theorem 0.4.7](#)). Hence  $|E| = |O| = n!/2$ .

**Exercise C2: Sign from the Number of Cycles.**

Suppose  $\sigma \in S_n$  is written as a product  $\sigma = \gamma_1\gamma_2 \cdots \gamma_c$  of **disjoint** cycles (no element appears in two of them) whose lengths  $k_1, \dots, k_c$  add up to  $n$ , where fixed points are included as cycles of length 1. (Such a product always exists: follow each element around its loop, as in [Example \(Computing with cycles\)](#).)

- (a) Prove that  $\text{sgn } \sigma = (-1)^{n-c}$ .
- (b) Use (a) to find the sign of  $\sigma = (1\ 4\ 6)(2\ 5) \in S_6$ , and compare with [Example \(A Sign Computed Two Ways\)](#).
- (c) Deduce that an  $n$ -cycle in  $S_n$  and the identity in  $S_n$  have the same sign exactly when  $n$  is odd.

*Solution.*

- (a) By [Theorem 6.3.1](#) and [Proposition 6.3.4](#),

$$\text{sgn } \sigma = \prod_{m=1}^c \text{sgn } \gamma_m = \prod_{m=1}^c (-1)^{k_m-1} = (-1)^{(k_1+\cdots+k_c)-c} = (-1)^{n-c}.$$

- (b) Including the fixed point 3 as the cycle (3),  $\sigma = (1\ 4\ 6)(2\ 5)(3)$ , with lengths  $3 + 2 + 1 = 6$  and  $c = 3$ . So  $\text{sgn } \sigma = (-1)^{6-3} = -1$ , as in [Example \(A Sign Computed Two Ways\)](#).
- (c) An  $n$ -cycle has  $c = 1$  and sign  $(-1)^{n-1}$ ; the identity consists of  $n$  cycles of length 1, so  $c = n$  and its sign is  $(-1)^0 = 1$ . These agree exactly when  $n - 1$  is even, that is, when  $n$  is odd.